

SOLUTION-MID EXAM II

Q1: Assume that, in a Stop-and-Wait ARQ system of a data communication network, the bandwidth of the line is 1 Mbps and 1 bit takes 20 ms to make a round trip. If each data frame has a length of 1000 bits, **analyze** the utilization percentage of the link. **[2.5 marks]**

Solution:

The bandwidth-delay product is

$$(1 \times 10^6) \times (20 \times 10^{-3}) = 20,000 \text{ bits}$$

The system can send 20,000 bits during the time it takes for the data to go from the sender to the receiver and then back again. However, the system sends only 1000 bits. We can say that the link utilization is only 1000/20,000, or 5 percent. For this reason, for a link with a high bandwidth or long delay, the use of Stop-and-Wait ARQ wastes the capacity of the link.

Q2: A Pure ALOHA network transmits 200-bit frames over a 200 kbps shared channel. If the vulnerable time increases by 25%, **analyze** the channel utilization (%) when the system collectively generates 1000 frames per second. **[2.5 marks]**

Solution:

Given:

- Frame size = 200 bits
- Channel rate = 200 kbps = 200,000 bps
- Frame generation rate = 1000 frames/sec
- Vulnerable time increases by 25%

Step 1: Frame transmission time

$T_f = \text{frame size} / \text{channel rate}$

$$T_f = 200 \text{ bits} / 200,000 \text{ bps} = 0.001 \text{ s} = 1 \text{ ms}$$

Step 2: Vulnerable time

Standard Pure ALOHA vulnerable time: $T_v = 2 T_f$

$$\text{With a 25\% increase: } T_v' = 1.25 \times 2 T_f = 2.5 T_f$$

Step 3: Traffic load G

$G = \text{generation rate} \times T_f$

$$G = 1000 \text{ frames/s} \times 0.001 \text{ s} = 1$$

Step 4: Throughput (normalized) for modified vulnerable time

Standard Pure ALOHA throughput: $S = G e^{-2G}$ (vulnerable time = $2T_f$)

Modified vulnerable time makes the exponent 2.5 instead of 2, so:

$$S' = G \times e^{-2.5 G}$$

Substitute $G = 1$:

$$S' = 1 \times e^{-2.5} \approx 0.08208499862$$

Step 5: Channel utilization (%)

$$\text{Utilization} = S' \times 100\% \approx 0.082085 \times 100\% \approx 8.21\%$$

Q3:

1. Propagation Time as a Fraction of Transmission Time:

Transmission time $T = 1000 \mu\text{s}$.

Propagation delay $\tau = 50 \mu\text{s}$.

Thus, $a = \tau/T = 50 / 1000 = 0.05$ (5%).

2. Throughput S for $G = 0.5$ in Non-Persistent CSMA:

Using the formula:

$$S = [G * e^{-aG}] / [G(1 + 2a) + e^{-aG}]$$

Substituting $a = 0.05$, $G = 0.5$:

$$e^{-0.025} \approx 0.9753099$$

$$S \approx 0.3197$$

3. Channel Utilization for $G = 0.5$:

$$\text{Utilization} = S \times 100\%$$

$$\approx 31.97\%.$$

4. Effect of Increasing Propagation Time (a):

- As a increases, the vulnerable period becomes longer.

- This increases collision probability.

- Throughput decreases because:

- e^{-aG} decreases
- The denominator $G(1 + 2a)$ grows

Thus, larger $a \rightarrow$ lower efficiency and more collisions